Design Basis Accident — Military Action

Energy Sector Series Introduction

An Introduction to the DBA-EN Framework and Document Architecture

Governing Parent: DBA-EN-SF — Energy Sector Design Basis Framework

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Preface

The DBA-MA Introduction — the companion document to this one, addressing the nuclear dimension of this work — begins in Ukraine, because that is where the gap it describes became undeniable. This document begins in the same place, because the energy infrastructure work that eventually became the DBA-EN series began there, too.

Over the course of nearly a decade and seven trips to Ukraine, I worked alongside their critical infrastructure professionals on grid resilience, cyber defense, incident response, and the coordination structures that hold together under pressure. What I came to understand — before the February 2022 invasion, before the Zaporizhzhia occupation, before the kinetic strikes on Ukrainian thermal plants and substations that followed — was that the analytical frameworks governing the protection of energy infrastructure in the United States and its allies had been built for accidents. They had been built for equipment failures, natural hazards, and, in some cases, deliberate sabotage by individuals or small groups. They had not been built for armies.

The evidence for that conclusion did not require the 2022 invasion. It had been accumulating for years. The December 2015 and December 2016 cyberattacks on the Ukrainian power grid — the first confirmed deliberate cyberattacks to cause civilian power outages anywhere in the world — demonstrated that a sophisticated adversary could reach inside a grid control system and turn off the lights. The 2019 Abqaiq-Khurais drone and cruise missile attack in Saudi Arabia demonstrated that precision standoff weapons could achieve immediate, large-scale disruption of petroleum production infrastructure with attribution designed for deniability. The 2022 Nord Stream pipeline sabotage demonstrated that state actors were willing and capable of destroying underwater energy infrastructure — critical to European energy security — through operations designed to be deniable and irreversible. The 2021 Colonial Pipeline ransomware event, while not a state military operation, demonstrated that the control systems of the largest petroleum product pipeline in the United States could be leveraged to produce a voluntary shutdown whose consequences rivaled a physical attack. Winter Storm Uri, in the same year, demonstrated how a compound gas-electric infrastructure failure could cascade beyond the ability of normal operational response to manage — a preview of what a deliberate adversary exploiting the same interdependency could achieve.

None of these events, individually, established that the design basis frameworks governing U.S. energy infrastructure were inadequate. Together, they established it beyond a reasonable doubt. The frameworks were built on the assumption that the initiating event — whatever it was — would be bounded in time, would not adapt in response to the operator's mitigation efforts, and would not actively prevent the response that the framework assumed would be available. Military action violates every one of those assumptions simultaneously. And yet, as of the time this series began, no formal methodology existed for analyzing the consequences of military action against an electric power plant, a natural gas processing facility, a petroleum refinery, or a

liquid natural gas terminal. The design basis frameworks for those facilities — NERC CIP, PHMSA integrity management, TSA security directives — were built to address reliability, safety, and terrorism. They were never designed to bind an adversary with the intentionality, persistence, precision, and denial capability of a state military force.

The DBA-EN series is the analytical response to that gap. It is built on the same methodological foundation as the DBA-MA nuclear series — the Design Basis Accident framework that has governed nuclear safety since its formalization in the 1960s and 1970s — applied with sector-specific precision across the electric, natural gas, and petroleum sub-sectors. It does not replace the existing regulatory frameworks. It extends them to bound the threat that no existing framework addresses.

There is one dimension of this series that I want to address directly before the document proceeds to its technical structure, because it is the dimension that I believe is most consequential for the next decade and the one that is most often treated as an afterthought in infrastructure security analysis. Every instrument in the DBA-EN series — every facility class document across all three sub-sectors — requires that the facility define a design basis for the AI and machine learning systems operating in its control plane. Not a cybersecurity assessment. Not a vendor questionnaire. A design basis: a formal analytical statement of what those systems must be capable of withstanding, what their boundaries of safe autonomous operation are, and where the line falls between automation that is permissible and automation that requires human oversight.

That requirement is new. No prior regulatory framework in the electric, gas, or petroleum sectors requires it. The DBA-EN series established it from the beginning — not reactively, after an AI-related incident revealed a gap, but proactively, because the analytical work on the physical design basis kept arriving at the same conclusion: the attack surface that matters most in the next decade is not the boiler, the pipeline, or the power transformer. It is the control plane. And the control plane is increasingly occupied by systems whose behavior under adversarial conditions has never been formally bounded.

This series is my attempt to close that gap before the adversary exploits it.

1. Purpose of This Document

The DBA-MA Introduction established the conceptual foundation for the Design Basis Accident — Military Action framework: what design bases are, why the existing ones are inadequate for military action, and how the DBA-MA series addresses that inadequacy in the nuclear sector. Readers of this document are assumed to have that foundation. This document does not reconstruct it. It builds on it.

The purpose of this document is to provide, for the DBA-EN Energy sector series, the same orientation that the DBA-MA Introduction provides for the nuclear series: the intellectual framework, the analytical structure, and the governing contributions of the work before the

reader engages the technical instruments themselves. It is written for the technically sophisticated reader — the energy engineer, the grid reliability analyst, the pipeline security professional, the federal regulator, the defense planner — who needs to understand what the series is for and why it is structured as it is before engaging with any individual instrument.

The document proceeds in five steps. First, it describes how energy infrastructure is currently protected — the existing regulatory frameworks and what they do and do not bind. Second, it identifies precisely where those frameworks break down under military action conditions, using the same four-characteristic analysis that the DBA-MA Introduction applied to the nuclear sector. Third, it describes the DBA-EN series architecture and how it closes the gap. Fourth, it addresses the AI governance design basis as the series's most consequential analytical contribution. Fifth, it provides the document inventory across all three sub-sectors.

2. How Energy Infrastructure Currently Stays Safe

2.1 The Existing Regulatory Architecture

The protection of U.S. energy infrastructure rests on a layered regulatory architecture built up over decades through a combination of mandatory standards, voluntary guidelines, and industry best practices. The architecture is sophisticated, technically grounded, and genuinely effective against the threats it was designed to address. Understanding what it does and does not bound is the essential prerequisite for understanding what the DBA-EN series adds.

The electric sector's primary protective framework is the North American Electric Reliability Corporation's Critical Infrastructure Protection (NERC CIP) standards. CIP establishes mandatory requirements for the cybersecurity of bulk electric system assets — electronic security perimeters, access controls, incident reporting, recovery planning, and, through CIP-014, physical security for transmission substations meeting defined criticality criteria. CIP is a genuine security standard with teeth: violations produce financial penalties, public reporting, and regulatory scrutiny. It is, in the global context of energy infrastructure regulation, among the most rigorous mandatory frameworks in existence.

The natural gas sector's primary protective framework is a combination of PHMSA's pipeline safety regulations (49 CFR Part 192 and Part 195), TSA's Pipeline Security Directives (issued following the Colonial Pipeline event), and FERC's reliability and market oversight for interstate pipeline operations. PHMSA's integrity management framework requires operators to assess and mitigate risks at high-consequence areas — populated zones, environmentally sensitive areas, drinking water sources — and to demonstrate that their pipelines can withstand foreseeable operating and environmental stress conditions. TSA's cybersecurity directives established the first mandatory framework for pipeline control system security. Together, they constitute a reasonably complete reliability and safety framework.

The petroleum sector's framework is distributed across EPA's Spill Prevention, Control, and Countermeasure (SPCC) regulations, OSHA's Process Safety Management standard, EPA's Risk Management Program, BSEE's offshore safety regulations, and TSA's voluntary pipeline security guidelines for petroleum pipelines. The petroleum sector has no mandatory security standard equivalent to NERC CIP-014 for its highest-consequence facilities.

2.2 What These Frameworks Bound

The existing frameworks share a common analytical structure: they define the credible threats against which protective measures must perform, they establish the required protective measures, and they specify the verification requirements that confirm the measures are in place and effective. The credible threats they were designed to bound are, without exception, in one of three categories: equipment failure and operational error, natural hazards (extreme weather, seismic events, flooding), and deliberate attack by individuals or small groups — sabotage, vandalism, and terrorism. Each of these threat categories has a defining characteristic that the frameworks' analytical approach implicitly relies upon: the threat does not adapt in response to the facility's protective measures.

An equipment failure does not observe the operator's response and redoubles its attack. A hurricane does not redirect toward a hardened facility and away from a vulnerable one. A saboteur who is deterred by a perimeter fence moves on. The regulatory frameworks built around these threat categories are built on the assumption that adequate protective measures, properly verified, will perform reliably because the threat they face is passive or bounded in its adaptation. That assumption is the load-bearing element of the entire framework. It is also the assumption that military action destroys.

2.3 The Design Basis Concept Applied to Energy Infrastructure

The nuclear sector developed the Design Basis Accident framework — the formal methodology for defining the envelope of postulated events that a facility must be demonstrated to withstand — in the 1960s and 1970s. The energy sector never developed an equivalent. NERC CIP's reliability standards address operational reliability; they do not establish a formal consequence envelope for facility-level physical events. PHMSA's integrity management addresses pipeline safety at high-consequence areas; it does not establish a bounding event methodology equivalent to the nuclear DBA. The TSA pipeline security directives address control system cybersecurity; they do not bound the physical consequences of a coordinated kinetic and cyber attack.

What the energy sector has, instead of a formal DBA framework, is a collection of planning standards that address specific foreseeable disruptions: N-1 and N-1-1 planning criteria for the electric transmission system, worst-case discharge scenarios for petroleum pipelines, and maximum credible spill volumes for marine terminals. Each of these is a planning standard that serves a specific regulatory purpose. None of them constitutes a formal design basis in the nuclear sense — a comprehensive analytical statement of the bounding events a facility must

withstand, the consequence envelope those events define, and the acceptance criteria that determine whether the facility's protective measures are adequate. The DBA-EN series supplies that missing framework.

3. The Gap: Where the Frameworks Break Down

3.1 The Four Characteristics of Military Action

The DBA-MA Introduction identified four characteristics of military action that the traditional design basis frameworks — in both the nuclear and non-nuclear sectors — were never designed to address. Those same four characteristics apply with full force across energy infrastructure, producing gaps that are in some cases more severe than those in the nuclear sector.

Intentionality. The adversary chooses what to attack, when, and in what sequence. In the energy sector, this means target selection informed by detailed grid topology analysis, pipeline flow modeling, and facility engineering data — much of which is available in public-domain NERC, FERC, and PHMSA filings. An adversary who understands grid operations can identify which transmission substations provide local voltage support in constrained load pockets, which compressor stations are hydraulic chokepoints in gas supply corridors, which processing plants would simultaneously curtail gas supply and petrochemical feedstock if destroyed, and which salt cavern storage fields provide the high-deliverability sprint capacity the grid needs during cold-snap emergencies. The protection frameworks built around these facilities were not designed against an adversary who has performed that analysis.

Persistence. The threat does not resolve on its own timeline. The adversary who destroys the Appalachian coal plant turbine rotor does not stop being an adversary while the 22-to-30-month replacement is manufactured. The adversary who compromises the pipeline SCADA system before destroying the pump stations does not go away during the 5-to-8-month recovery. Persistence means that every phase of the recovery — component procurement, contractor mobilization, regulatory restart authorization — is subject to adversarial interdiction. The NERC CIP framework addresses cybersecurity. It does not address an adversary who interdicts the supply chain for the replacement components required by a CIP incident's physical recovery.

Precision. Modern adversaries can target specific systems. The coal plant operator whose steam turbine rotor is selected as the governing long-lead-time component in an RT_HVLLC calculation, the refinery operator whose HF alkylation unit is selected for its off-site toxic consequence potential, the solar farm operator whose collector substation is the chokepoint through which 500 MW of generation flows — none of these facilities' protection frameworks was designed against an adversary who has read their facility engineering documentation and selected the specific component whose destruction produces the maximum consequence. NERC CIP-014 requires vulnerability assessments for high-consequence transmission substations; no equivalent requirement exists for most energy infrastructure.

Denial. Military action can prevent the mitigation response. This is the characteristic with the most varied and consequential expression across the energy sector. The re-pressurization of a depressurized urban gas distribution system requires technicians to visit every customer service connection — and those technicians cannot do their work if they cannot safely access the affected neighborhoods. The repair of a deepwater offshore pipeline requires specialized construction vessels that cannot operate in contested maritime environments or high-seas states. The replacement of a destroyed pipeline pump station requires specialty contractors who cannot mobilize to a contested area. The existing protection frameworks assume that the mitigation response is available. Military action removes that assumption.

3.2 The Evidentiary Record

The four characteristics are not theoretical. The evidentiary record across the decade prior to this series demonstrates each of them in documented operational events against energy infrastructure.

The 2015 and 2016 Ukrainian grid attacks demonstrated intentionality and precision applied to grid control systems: attackers pre-positioned in SCADA networks, selected specific circuit breakers for remote operation, coordinated the timing across multiple distribution companies, and used telephone denial-of-service attacks against utility call centers to delay the human response. The consequence — simultaneous outages affecting hundreds of thousands of customers in winter — was designed, not incidental.

The 2019 Abqaiq-Khurais attack demonstrated precision and persistence applied to petroleum production infrastructure: cruise missiles and drones targeted the specific stabilization columns and gas-oil separation trains at Saudi Aramco's largest facility, whose damage would produce the greatest immediate output reduction. The attack was designed for deniability — attribution was contested for weeks — demonstrating the denial of the political response as well as the physical one.

The 2022 Nord Stream pipeline sabotage demonstrated persistence and denial applied to subsea infrastructure: the physical destruction of the pipelines was irreversible given the political and economic context, producing a permanent loss of capacity whose consequence — European energy security degraded, German industrial competitiveness structurally impaired — extends years beyond the event. The subsea attack vector was designed to deny the attribution response and to make the repair operationally deniable.

The 2021 Colonial Pipeline event, while not a state military operation, demonstrated the voluntary shutdown dynamic that is the governing vulnerability of the pipeline facility class: a control system compromise sufficiently threatening to produce a voluntary operational shutdown whose consequences — Eastern Seaboard fuel shortages within days — were indistinguishable in their immediate effect from physical pipeline destruction. A state actor with the same objective and no constraint on kinetic action can produce the same supply disruption with a physical repair timeline measured in months.

The 2021 Winter Storm Uri demonstrated the gas-electric cascade: simultaneous gas supply curtailment and high heating demand, producing a compound infrastructure failure that killed hundreds of people, exceeded the operational response capacity of multiple state and federal emergency management systems, and demonstrated exactly the interdependency that a deliberate adversary can exploit with far greater precision than a weather event.

Each of these events fell partly outside the design basis of the affected infrastructure. Together, they define the design basis that was missing.

4. The DBA-EN Series Architecture

4.1 The Three-Layer Progression

The DBA-EN series applies the same three-layer analytical progression that the DBA-MA nuclear series established: Design Basis (DB), Design Basis Accident (DBA), and Military Action Extension (DBA-MA). The three layers are not alternatives; they are sequential. The DB layer establishes the postulated conditions — equipment failures, natural hazards, operational events, cross-sector cascades — that the facility must be demonstrated to withstand, cause-agnostic. The DBA layer defines the non-adversarial consequence envelope: the bounding events, their consequence taxonomy, and the acceptance criteria. The DBA-MA layer applies the four characteristics of military action to each postulated event category, building the adversarial extension on the DBA envelope rather than beside it.

This three-layer structure means that the DBA-EN series instruments are simultaneously the most rigorous non-adversarial consequence analyses of their facility types and the governing documents for military-action consequence assessment. The adversarial analysis does not exist separately from the safety analysis; it is built on it, extends it, and makes it complete.

4.2 The Three Sub-Sector Series

The DBA-EN series comprises three sub-sector series organized by the primary consequence chain of each facility type, not by NIPP administrative sector boundaries. The consequence-chain organizing principle means that a facility belongs in the sub-sector series whose primary consequence chain it propagates through — the series whose consequences it most directly produces when it fails. This principle was established in the DBA-EN-SF parent document and governs every instrument in the series.

DBA-ES — Electric Sector. Governs generation, transmission, distribution, and control of the company infrastructure whose primary consequence chain runs through the bulk electric system. This includes not only the conventional electric utility infrastructure but natural gas compressor stations in GenCo supply corridors (DBA-ES-GC-FC4), whose governing consequence is the gas-electric cascade, and the full GenCo generation fleet — coal, oil-fired, and renewable generation — whose consequence chains run through grid reliability rather than fuel supply.

DBA-GAS — Natural Gas Sector. Governs natural gas infrastructure whose primary consequence chain runs through the gas supply system: transmission pipelines and compressor stations (FC1), processing plants (FC2), underground storage (FC3), and local distribution (FC4). The LNG use case (UC-LNG) and the policy companion (POL-Deterrence) round out the series. The DBA-GAS instruments also govern the gas-electric cascade from the gas sector's perspective, complementing the DBA-ES analysis at the electric sector's boundary.

DBA-OIL — Petroleum Sector. Governs petroleum infrastructure whose primary consequence chain runs through the liquid fuel supply system: refining (FC1), offshore crude production (FC2), terminals and storage (FC3), and crude and product transport pipelines (FC4). The DBA-OIL series established three new facility class consequence frameworks — offshore production, terminal and storage, and pipeline transport — that had no formal design basis equivalent before this series.

4.3 The Consequence Taxonomy

Every instrument in the DBA-EN series applies a three-tier consequence taxonomy inherited from the DBA-EN-SF parent document. Tier 1 consequences are facility-level: consequences that are severe at the facility but do not propagate to the grid, pipeline system, or regional economy. Tier 2 consequences are system-level: consequences that affect the bulk power system, the interstate gas transmission system, or the regional petroleum supply network. Tier 3 consequences are cross-sector cascading: consequences that reach a critical function in another sector — the gas-electric cascade, the water sector dependency, the military logistics supply chain, the petrochemical feedstock system.

The three-tier taxonomy is the series' primary instrument for ensuring that facility-class consequence analysis is not confined to the facility. The DBA-EN series was built on the recognition that the most consequential energy infrastructure failures in the evidentiary record were not Tier 1 events — they were the compound, cascading failures whose severity was produced by the interaction between sectors. The consequence taxonomy formalizes that recognition as an analytical requirement.

4.4 The Cross-Sector Consequence Elevation Rule

A formal coordination protocol governs the boundary between the electric and gas sub-sector series at dual-scope facilities — compressor stations in GenCo supply corridors that are simultaneously within the scope of DBA-ES-GC-FC4 and DBA-GAS-FC1. The rule, established in DBA-EN-SF §3.4, is both analytical and architectural: each instrument analyzes from its own consequence chain perspective; the acceptance criteria of each instrument govern their own consequence tier; and a DBA-EN energy asset that is the sole supply path for a Tier 2 or Tier 3 DBA-designated facility in any sector must carry tightened acceptance criteria for that supply relationship. This rule is modeled on the European EPCIP/NIS2 critical infrastructure interdependency framework and fills a gap that NERC CIP standards do not address.

4.5 Governing Consequence Metrics

The DBA-EN series established consequence-duration metrics as series standards, filling gaps where no prior regulatory framework had defined a formal metric. Four are described here; DBA-EN-SF §4.4 states the full set.

RT_HVLLC (Recovery Timeline for High-Value Long-Lead Components). The time to restore a facility following destruction of its most difficult-to-replace component, expressed in months. RT_HVLLC is the governing metric wherever a physical attack's consequence is dominated by the procurement and replacement timeline of a specific critical component. In DBA-ES, it governs the large power transformer (18–24 months), the HVDC converter transformer (12–18 months), the coal steam turbine rotor (22–30 months), the combustion turbine hot section (8–14 months standard frames), and the offshore wind nacelle (12–24 months). The metric originated in the GenMix Optimization Whitepaper and is the first formally defined recovery timeline metric for generation infrastructure in the DBA-ES series.

GSD (Generation Supply Duration). The continuous period of full gas-fired generation curtailment due to insufficient pipeline pressure is measured in hours or days. GSD governs DBA-ES-GC-FC4 analysis of compressor station loss in GenCo supply corridors. It defines the governing metric for the gas-electric cascade from the electric sector's analytical perspective.

PPSD (Pipeline Pressure Supply Duration). Introduced in DBA-GAS-FC2 for processing plant outage consequence measurement. Governs the downstream gas supply as a consequence of a processing plant forced outage.

TCI (Transfer Capability Isolation). The duration of cross-interconnection transfer capability unavailability following HVDC link loss. Governs DBA-ES-TX-FC3.

5. The AI Governance Design Basis — The Series' Most Consequential Contribution

5.1 What Is New

Every instrument in the DBA-EN series — every facility class document across all three sub-sectors — requires that the facility define a design basis for the AI and machine learning systems operating in its control plane. This is analytically new. No prior regulatory framework in the electric, gas, or petroleum sectors imposes this requirement. NERC CIP addresses control system cybersecurity — the integrity and access controls around automated systems — but does not require the operator to define what those systems must be capable of withstanding under adversarial conditions, or to establish formal bounds on their autonomous operation. PHMSA's integrity management framework addresses pipeline safety; it does not address the behavior of ML-driven pipeline scheduling systems under adversarial input. TSA's Pipeline Security Directives address network segmentation and incident response; they do not address the AI

governance boundary between autonomous operation and human oversight in a SCADA environment.

The DBA-EN series established this requirement from the beginning — not as a response to an AI-related incident, but as a proactive architectural element built into the series from its inception. The reason is straightforward: the analytical work on the physical design basis for energy infrastructure kept arriving at the same conclusion. The attack surfaces that matter most in the next decade are not primarily physical. They are in the control plane. And the control plane is increasingly occupied by systems — advanced process control, real-time optimization, automated dispatch, inventory management, pressure regulation — whose behavior under adversarial conditions has never been formally bounded by any regulatory framework.

5.2 The Bright Line and Command Broker Architecture

The governing architectural principle for AI governance across the DBA-EN series is the Bright Line as defined in FD-BL (Foundational Definition — The Bright Line). Under FD-BL §4.2–§4.3 the Bright Line is a threshold on individual actions: an action affecting a physical process parameter is above the line when its incorrect performance could produce a consequence at or above the facility's severity threshold — requiring a human Command Broker enforced architecturally (FD-BL §4.5) — and below it when it could not. Placement follows the consequence of the action, not the technology class of the component; determinism is the scope carve-out of FD-BL §4.4, not the placement test. The Bright Line is not a performance standard or a cybersecurity control. It is an architectural requirement: the system must be designed so that the boundary is enforced structurally, not procedurally. This introduction does not restate the definition for each instrument; every DBA-EN instrument inherits the Bright Line by reference to FD-BL and develops only its facility-class-specific application.

The Command Broker is the human authority in the control loop for every above-Bright-Line action. The Command Broker requirement is not a review step added to an existing workflow. It is an architectural constraint: the system must be physically incapable of executing an above-line action without a positive human confirmation that is local, auditable, and not itself executable by a software agent. The Bright Line and Command Broker architecture was first applied in the series in the DBA-DC (hyperscale data center) instrument. The Bright Line is now defined in FD-BL, and the Command Broker architecture is stated in the Command Broker family of instruments, registered in CB-REG. Every DBA-EN instrument inherits both by reference. Its application is facility-class-specific in every instrument: each document identifies the AI governance interaction points for its facility type and places the actions at each on their consequence.

5.3 Facility-Class-Specific Contributions

The DBA-EN series made three analytical contributions to AI governance that are first-of-kind across the entire DBA series and that have implications beyond energy infrastructure.

The optimization-safety boundary (DBA-GAS-FC2). In continuous-process optimization facilities — gas processing plants, petroleum refineries, and any facility where APC and RTO systems operate — the Bright Line applies to the facility's own commercial optimization logic as well as to adversarial manipulation: optimization actions are placed on their consequence like any others; the region set back from every SIS trip point is a pre-authorized envelope (FD-EV §4.1), inside which an above-line action may be discharged in advance only through a verified gatekeeper under FD-BL-D1, and any movement of that envelope toward a trip point is above the line under any reading and excluded from advance discharge (FD-BL-D1 R3). This is a non-adversarial structural risk: an APC system continuously pushing process conditions toward the limits that maximize NGL recovery or product yield is simultaneously pushing those conditions toward the limits where safety systems are most likely to be tested. The AI Governance Design Basis must establish that APC operating envelopes include explicit safety margin setbacks from SIS trip points, enforced by architecture. DBA-GAS-FC2 was the first instrument in any sector's design basis framework to name and formalize this non-adversarial AI risk as a Design Basis requirement. It has been applied to DBA-OIL-FC1 (petroleum refining) and carries through to every processing facility in the series.

The inventory depletion attack surface (DBA-GAS-FC3, DBA-ES-GN-FC7). ML systems governing storage inventory management and automated dispatch may fall outside the Command Broker obligation under the FD-LD carve-out, where they are deterministic within a stated input bound and formally testable as deployed — a finding on scope, not a placement below the line — but they are simultaneously the attack surface for the most insidious adversarial scenario in their respective facility classes: the systematic, undetectable depletion of a resource buffer before the emergency that requires it. DBA-GAS-FC3 identified the withdrawal scheduling ML system as the mechanism by which underground storage inventory could be depleted months before peak heating season, without any physical attack and without triggering anomaly detection. DBA-ES-GN-FC7 applied the same analysis to oil-fired peaker automated dispatch: spurious starts that burn down the on-site fuel tank before the grid emergency arrives. Both instruments establish that the AI Governance Design Basis must address cumulative operational pattern analysis — not just individual transaction authorization — as a security requirement.

Inverter firmware trustworthiness as a Bright Line precondition (DBA-ES-GN-FC8). The determinism carve-out (FD-LD §4.3) holds only where a component's behavior is deterministic within a stated input bound and formally testable as deployed. DBA-ES-GN-FC8 established that for inverter-based resources this requires an additional precondition: the firmware governing the inverter's behavior must itself be independently verified as free of undisclosed functionality before the device can rely on the carve-out. An inverter operating on unverified firmware — a real prospect, given Chinese manufacturer dominance of 80 to 85 percent of global utility inverter production and CISA's documented findings of undisclosed remote access functionality — cannot show that the input bound over which its behavior is deterministic holds as deployed. Its autonomous behavior cannot be trusted as deterministic, regardless of its apparent normal-operating-conditions performance. This precondition is the most consequential single AI

governance analytical contribution in the DBA-EN series: it applies to every inverter-based resource in all three sub-sectors, at a fleet scale whose replacement timeline is measured in years, not months.

5.4 Why This Was Not Done Before

The absence of a formal AI governance design basis from prior energy infrastructure regulatory frameworks is not a regulatory oversight. It reflects the timing of AI deployment relative to the regulatory cycle. When NERC CIP was developed, advanced process control was well-established, but large language models were not. When TSA's Pipeline Security Directives were issued, the primary concern was ransomware and network intrusion — threats to the IT layer, not to the behavior of ML optimization systems in the OT layer. When the existing PHMSA integrity management framework was developed, the concept of a Gen AI advisory tool recommending pipeline scheduling decisions had no operational reality.

The AI governance design basis addresses the infrastructure protection gap that opens when the regulatory cycle lags the deployment cycle — as it always does. The DBA-EN series established the formal requirement before the incidents that would eventually compel it. That is what a Design Basis framework is for.

6. Document Inventory

This inventory names the instruments of the series and says what each does. It does not state their versions. The current version of every series instrument is stated in DBA-EN-REG, the Energy Sector Series Register, which is authoritative for the series; the instruments of the Command Broker family are registered in CB-REG. A version stated here would go stale at the next increment of any instrument, as the inventories of earlier versions of this document did.

6.1 Parent and Series-Wide Instruments

Document	Description
DBA-EN-SF: Energy Sector Design Basis Framework	Parent document governing all three sub-sector series. Establishes the DB→DBA→DBA-MA three-layer progression, the consequence-chain organizing principle, the gas-electric bidirectional cascade as a first-class cross-sector postulated event, the cross-sector consequence elevation rule, the dual-DB architecture, and the Bright Line and Command Broker requirements as series-wide standards.
DBA-EN-REG: Energy Sector Series Register	The authoritative version state of the series, so that citations to its instruments need not carry version tokens. Took that role from DBA-EN-SF §1.4.
DBA-EN-UC-1: Gas-Electric Cascade	Cross-sector use case. A full-loop trace of the bidirectional gas-electric cascade from initiating event to terminal consequence state, from both sub-sectors' perspectives.

DBA-EN-UC-2: Coordinated Multi-Node Attack	Cross-sector use case. Minimum target set analysis for an interconnection: how an adversary defeats N-1 recovery by design.
DBA-VC: Vendor Companion	Cross-series vendor companion. Translates the series' AI governance requirements into vendor product-conformance obligations and the artifacts an asset owner needs from a supplier.
CB-UC-1: BA Integrated Interface	The Command Broker family's electric-sector application. Per-action Bright Line placement for Balancing Authority operations and the integrated confirmation-interface specification.

6.2 DBA-ES — Electric Sector

Document	Description
DBA-ES Sector Framework	Electric sub-sector parent. Governs all DBA-ES facility-class instruments. Establishes the consequence-chain organizing principle, the GenCo/TransCo/DisCo/CntlCo taxonomy, the RT_HVLLC metric, and the Bright Line and Command Broker requirements for the electric sector.
DBA-ES-HY-FC1	Large hydroelectric storage (>500 MW). Governing consequence chains: bulk generation loss and downstream dam failure flood wave. RT_HVLLC: major penstock and turbine components.
DBA-ES-TX-FC2	Large EHV transmission substation (345–500 kV, metropolitan ring). Governing RT_HVLLC: large power transformer, 18–24 months. NERC CIP-014 reference facility class.
DBA-ES-TX-FC3	HVDC interconnect (>2,000 MW, cross-interconnection). Governing metric: TCI. RT_HVLLC: converter transformer, 12–18 months.
DBA-ES-GC-FC4	Major natural gas compressor station in the GenCo supply corridor. Governing metric: GSD. Placed in DBA-ES because the primary consequence chain runs through the gas-electric cascade.
DBA-ES-DI-FC5	Distribution substation with edge generation (islanding capability). Full three-layer instrument.
DBA-ES-CntlCo	Balancing Authority / Reliability Coordinator / Transmission Operator coordination class. Full three-layer instrument; Situational Awareness Blackout Duration (SABD) is its governing consequence metric.
DBA-DC (Critical Loads grouping)	Hyperscale data center (>100 MW IT load) and the Critical Loads grouping within the electric sub-sector, whose eleven members are recorded individually in DBA-EN-REG. The DBA-DC framework is where the Bright Line and Command Broker architecture were first applied in the series.
DBA-ES-GN-FC6	Coal-fired generation (>750 MW or must-run/transmission-essential). Governing RT_HVLLC: supercritical steam turbine rotor, 22–30 months — longest in the generation series. Establishes coal supply interdiction (Scenario B) and CCR impoundment environmental cascade (Scenario C).
DBA-ES-GN-FC7	Oil-fired generation and peakers (>100 MW or RMR designation). Governing RT_HVLLC: combustion turbine hot section, 8–14 months standard frames. Establishes the backstop removal attack (Scenario A) and fuel inventory depletion through spurious automated dispatch (Scenario B).

DBA-ES-GN-FC8	Renewable generation and BESS — utility-scale solar, wind (onshore and offshore), BESS. Establishes the inverter firmware attack (Scenario A) as the first supply-chain-primary attack scenario in the DBA-EN series. Introduces inverter firmware trustworthiness as a Bright Line precondition.
DBA-ES-MA-SF	Electric sector-level Military Action companion (Sector Framework). Characterizes the MA threat at the sub-sector level, the four-filter applicability gate, common attack surfaces, the pattern attack framework, and Tier A consequence analysis. Companion layer to the DBA-ES facility-class instruments.
DBA-ES-MA-QB	Electric Sector MA Quantitative Design Basis Homes the cross-FC quantitative MA content that no single instrument can carry: SABD series-level calibration bounds, pattern-attack restoration stress-test, and sector-level concurrent spare-equipment demand.
DBA-ES-POL: Mandatory Standards	Policy companion. The mandatory standards imperative: extending NERC CIP-014 to the full DBA-ES consequence envelope.

6.3 DBA-GAS — Natural Gas Sector

Document	Description
DBA-GAS Sector Framework	Natural gas sub-sector parent. Governs all DBA-GAS facility-class instruments.
DBA-GAS-FC1	Natural gas transmission — interstate pipeline and compressor stations. The most mature instrument in the DBA-GAS series. Establishes the gas-electric cascade from the gas sector's perspective.
DBA-GAS-FC2	Natural gas processing. First instrument in the DBA-EN series to establish the optimization-safety boundary as a formal Design Basis requirement. Dual consequence chain: gas supply curtailment and NGL petrochemical feedstock disruption simultaneously.
DBA-GAS-FC3	Underground storage — depleted reservoir and salt cavern. Establishes the sprint/endurance distinction. First instrument in the DBA-EN series to establish inventory depletion through ML scheduling manipulation as a governing DBA-MA scenario.
DBA-GAS-FC4	Local distribution — LDC city gate to end-use. Governs direct residential life-safety consequences. Establishes the re-pressurization workforce constraint as the governing recovery metric unique to this facility class. Resolves OI-4 (DBA-EN-SF).
DBA-GAS-FC-LNG	LNG terminals — liquefaction, storage, and marine transfer. Promoted from use-case status (DBA-GAS-UC-LNG retained as the bounding-event annex) on strategic significance: allied export commitments and a distinct marine attack surface warranting a recurring design basis.
DBA-GAS-MA-SF	Natural gas sub-sector Military Action companion (Sector Framework). Mirrors DBA-ES-MA-SF; Tier A covers transmission and compression, LNG terminals, processing, and gas control.
DBA-GAS-UC-LNG	LNG import and export terminal use case.
DBA-GAS-POL-Deterrence	Policy companion — deterrence framework for natural gas infrastructure.

DBA-MA-ONG Series Introduction	Oil and natural gas series introduction — predecessor orientation document for the DBA-GAS and DBA-OIL series.

6.4 DBA-OIL — Petroleum Sector

Document	Description
DBA-OIL Sector Framework	Petroleum sub-sector parent. Governs all DBA-OIL facility-class instruments.
DBA-OIL-FC1	Petroleum refining. The most mature petroleum sector instrument. Establishes the multi-product consequence chain and the fired heater as the governing DBA-MA kinetic target.
DBA-OIL-FC2	Offshore crude production. Governs the maritime threat profile and remote access constraints. Establishes the offshore construction vessel as the governing recovery constraint — the offshore analog of the pipeline pump unit strategic spare gap.
DBA-OIL-FC3	Crude and product terminals and storage. Establishes the terminal as the logistics bridge. Scenario C (Northeast heating oil terminal attack — seasonal timing as force multiplier) is the most consequential seasonal consequence envelope in the DBA-OIL series. Strategic marine terminals have been carved out to DBA-OIL-FC5; FC3 now covers general onshore and non-strategic marine terminals and storage.
DBA-OIL-FC4	Crude and product transport — trunk and product pipelines. Establishes the voluntary shutdown problem as the governing vulnerability for the pipeline facility class. The Colonial Pipeline template (Scenario A) with kinetic amplification and the Hormuz closure compound timing (Scenario C) define the bounding consequence envelope.
DBA-OIL-FC5	Marine crude and product terminals — strategic import/export. Carved from FC3 on the strategic boundary: a terminal is FC5 when it carries a defense-fuel logistics commitment, an allied or strategic export commitment, or is a designated strategic crude-import chokepoint.
DBA-OIL-MA-SF	Petroleum sub-sector Military Action companion (Sector Framework). Mirrors DBA-ES-MA-SF; Tier A covers refining, marine terminals, transport, and offshore production; control is treated as a cross-cutting attack surface.

6.5 Companion and Supporting Documents

Document	Description
GenMix Optimization Whitepaper	Origin document for the RT_HVLLC metric and Table 1 RT_HVLLC values. Foundational reference for DBA-ES-GN-FC6/FC7/FC8.
Predictive UFLS Ride-Through Position Paper	PMU and under-frequency load shedding analysis. Grid stability reference for DBA-ES.
DBA-UxS Cross-Sector	Uncrewed systems threat framework — cross-sector foundational document.
DBA-MA-UxS-NA	Nuclear implementation annex to DBA-UxS.
GenAI Risk ICS	GenAI risk in industrial control systems. Foundational reference for AI governance design basis across the DBA-EN series.

Command Broker Implementation Guide	Implementation guidance for the Bright Line and Command Broker architecture. Registered in CB-REG.
CB-Framework Cross-Sector	Command Broker family parent. Establishes the broker requirement, the criticality-band framework, and the Command Broker / Influence Broker distinction. Registered in CB-REG.
CB-IB Qualification Standard	Command Broker / Influence Broker qualification and onboarding standard. Registered in CB-REG.
UA Grid Defense Cyber-Kinetic	Ukrainian grid defense under cyber-kinetic attack — primary evidentiary anchor for DBA-ES kinetic attack scenarios.
Powering the Always-On Economy	Energy sector policy context document.
Fuel Cell Position Paper	Fuel cell generation technology position.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial version. Companion orientation document to the DBA-MA Introduction (v1.9), establishing the conceptual foundation, analytical structure, and document inventory for the DBA-EN Energy sector series. Covers all three sub-sector series: DBA-ES (Electric, sector framework v0.8, eleven instruments), DBA-GAS (Natural Gas, sector framework v1.6, four FC instruments and two companions), and DBA-OIL (Petroleum, sector framework v1.6, four FC instruments). Establishes the AI governance design basis as the series's most consequential analytical contribution. Documents three series-first AI governance contributions: the optimization-safety boundary (DBA-GAS-FC2), the inventory depletion attack surface (DBA-GAS-FC3 / DBA-ES-GN-FC7), and inverter firmware trustworthiness as a Bright Line precondition (DBA-ES-GN-FC8).
0.2	June 2026	T. Roxey	Parent-registration pass. §6.1 parent version corrected to v0.9. §6.2 DBA-ES: DBA-ES-MA-SF (v0.2) and DBA-ES-MA-QB (v0.1) registered; Sector Framework corrected to v0.9. §6.3 DBA-GAS: DBA-GAS-FC-LNG and DBA-GAS-MA-SF (v0.2) registered; Sector Framework to v1.7. §6.4 DBA-OIL: DBA-OIL-FC5 and DBA-OIL-MA-SF (v0.1) registered, FC3 scope noted as split; Sector Framework to v1.8. Mirrors the registrations in DBA-EN-SF v0.9. Sub-sector FC instrument versions otherwise not refreshed (separate pass).
0.3	June 2026	T. Roxey	Series Introduction refresh pass. Instrument version numbers in §6.2 (DBA-ES), §6.3 (DBA-GAS), and §6.4 (DBA-OIL) reconciled against the authoritative DBA-EN-SF v0.9 §1.4 inventory: DBA-ES-HY-FC1→v0.5, DBA-ES-TX-FC2→v0.3, DBA-ES-TX-FC3→v0.3,

			DBA-ES-GC-FC4→v0.5, DBA-ES-DI-FC5→v0.4, DBA-ES-CntlCo→v0.3, DBA-DC→v1.7, DBA-ES-GN-FC6/FC7/FC8→v0.2; DBA-GAS-FC1→v1.6, DBA-GAS-FC2→v0.2, DBA-GAS-POL→v1.3; DBA-OIL-FC1→v1.3, DBA-OIL-FC4→v0.3. DBA-ES-POL (v0.1 in §1.4) remains absent from §6.2 and is flagged rather than added pending an accurate description. No content beyond version reconciliation changed.
0.4	September 2026	T. Roxey	§ AI-governance placements conformed under OI-R3-1 (Governing Instructions v0.17 §6) on the inherit-by-reference model. Placements previously asserted from the character of the component — deterministic, ML, or protective — are restated as the FD-LD scope carve-out applied to in-bound actions, with placement itself referred to DBA-EN-SF §4.2 rather than restated in this document. Restating the placement criterion per instrument is the drift mechanism that allowed the error to propagate; this revision removes the restatement. No physical Design Basis content, consequence threshold, event code, scenario, or metric is changed.
0.5	October 2026	T. Roxey	Inventory and conformance revision ahead of re-publication, on Project 0's split ruling of 27 September 2026. §6 no longer states versions: the Version column is removed from all five inventory tables, and a lead-in names DBA-EN-REG as the authoritative register for the series and CB-REG for the Command Broker family. The inventory had pinned versions months stale. §6.1 becomes Parent and Series-Wide Instruments and gains DBA-EN-REG, DBA-EN-UC-1, DBA-EN-UC-2, DBA-VC and CB-UC-1, the last moved from §6.5. §6.2 gains DBA-ES-POL. The DBA-DC row names the Critical Loads grouping. The DI-FC5 and CntlCo descriptions no longer say their DBA and MA layers are deferred, and CntlCo is described as a coordination class. Command Broker family descriptions use Influence Broker and the criticality-band framework. In §5, the Command Broker paragraph no longer ties the broker to Gen AI by technology, and the optimization-safety, inventory-depletion and firmware paragraphs state the carve-out as a finding on scope rather than a placement below the line (FD-BL §4.2, §4.4). §4.5 no longer says the series has four metrics.
0.6	October 2026	T. Roxey	Wording conformance to the October 2026 Edition of the Foundational Definitions, on Project 0's work order of 3

			October 2026 (drift-screen finding P8-09). The inverter-firmware paragraph in Section 4 cites the determinism carve-out to FD-LD §4.3 in place of FD-BL §4.4. Found in the same reading under CR-VA and conformed with it: the optimization-safety boundary paragraph placed optimization actions inside the setback region below the line because they were inside it, which is placement by envelope; it now places them on consequence, names the setback region as the pre-authorized envelope it is (FD-EV §4.1), and states that movement of the envelope toward a trip point is above the line and excluded from advance discharge (FD-BL-D1 R3). No other change. Publication handling under the CR-PB draft: v0.5 is current on the publications site, so this pass acquires a republish step. Revision-history entries were excluded from every substitution and read afterwards to confirm the exclusion held.

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